

AI Enemy Design

Philosophy

- Enemies should feel intentional, not deterministic
- Every action should make sense given the current situation, but there should still be enough variety that players can't perfectly predict what comes next
- **Proactive Actions**: Players perceive reasonable variance as intelligence
- **Reactive Actions**: Players perceive mostly correctness as intelligence

Proactive Actions

- Example: Attack, Strafe, Flank, etc.

Scoring System

- **Uniform Randomness**
- Filter out actions that are “bad”
- Choose completely randomly from the “good” ones
- Reason: Increases behavioral variety and makes enemies feel less scripted while keeping them from making weird actions

Reactive Actions

- Example: Block, Dodge, Parry, etc.

Scoring System

- **Weighted Randomness**
- Filter out actions that are “bad”
- Roll a result from the “good” ones Roulette Style where each action has a higher chance of being chosen the higher their score is
- Reason: AI still shows variety, but it naturally favors the strongest tactical reaction while allowing occasional alternatives

Why not use weighted randomness for everything?

Weighted randomness on proactive actions tends to over-favor the highest-scoring move, so enemies can become predictable and repeatedly use the same attack pattern. Uniform selection among viable actions produces more varied encounters. In contrast, reactive decisions should

appear intelligent and context-aware, so weighting by score helps the AI choose the most appropriate defensive response more consistently.

Interruption Rules

- Actions define when they can be interrupted
- Reactive sequence will only try and execute when something prompts it to and the current action can be interrupted
 - EX: Charging up an attack → Attack Detected → Cancel into dodge
- Proactive sequences never interrupt each other

Why not let proactive sequences interrupt each other?

Proactive actions interrupting each other tend to look robotic and unnatural. If an enemy is far away and about to shoot a projectile, it shouldn't all of a sudden cancel its current attack and start a sword swing just because their target got closer.

It makes more sense and looks better if an enemy stops an attack charge-up with a dodge because they're about to get hit, especially if for example that attack would kill them.